

JKBOSE 2026

CLASS 11TH

Anatomy Of Flowering Plants

LECTURE-01

BOTANY

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Topics To Covered

- 1** Introduction
- 2** OVERVIEW OF PLANT TISSUE
- 3** TISSUE SYSTEM

INTRODUCTION

- Plant Anatomy → Study of internal Structures of a plant.
 - Internal Structure of a plant shows adaptations towards their Environment.
 - Monocots and Dicots } of Angiosperms are anatomically different.

OVERVIEW OF PLANT TISSUE



TISSUE → Term "tissue" was coined "Grew".

→ Study of TISSUE is called as Histology.

→ TISSUE is a group of cells having common origin and usually performs common function.

NOTE → Cells of a tissue may be same / different.

Example

{ Same type of cells are present in nervous tissue.

{ Different type of cells are present in xylem, phloem.

TISSUE

On the basis of Division Capability

Meristematic Tissue

- Continuously dividing tissue.
- No Specific function. (Immature tissue)

03 types

Apical meristem

- ≡ Present in tip of Root and Stem
- ≡ Causes 1^o growth i.e. increase in length.
- Ex: RAM (Root Apical meristem)
- SAM (Shoot " ")

Intercalary meristem

- ≡ present between mature tissue (Near Nodes)
- ≡ present in Monocots.
- ≡ Cause 1^o growth.

Lateral meristem

- ≡ present along the lateral sides of plant.
- ≡ Causes 2^o growth in Dicots. i.e. increases width of plant.
- Ex: Vascular Cambium, Cork Cambium.

PERMANENT TISSUE

- Mature tissue (does not divide)
- Performs Specific functions.

02 types

Simple permanent tissue

- ≡ All the cells are of same type.
- Ex: Parenchyma, Collenchyma, Sclerenchyma.

Complex Permanent Tissue

- ≡ Composed of different type of cells.
- ≡ Xylem, phloem.

TISSUE SYSTEM

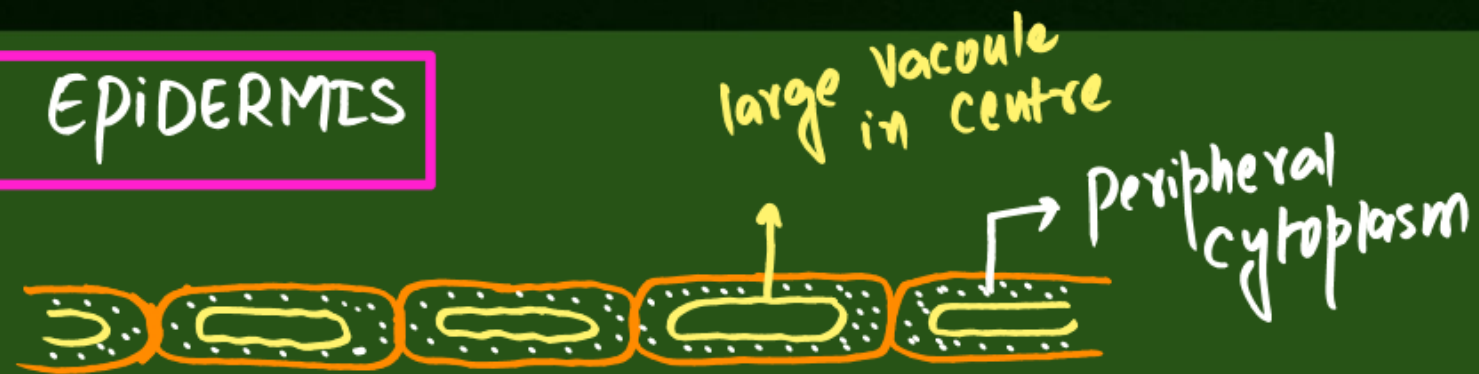


- ≡ In plants, tissues combine to form tissue system.
- ≡ Structure and function of a tissue depends on its location.
- ≡ On the basis of structure and location, tissue system is of 03 types:
 - a. Epidermal tissue system ★
 - b. Ground tissue system / Fundamental tissue system.
 - c. Vascular tissue system / Conducting tissue system.

a. Epidermal tissue System

- It forms the Outermost Covering of plant.
- It consists of:
 1. Epidermis (made of epidermal cells)
 2. Epidermal Structures (ex. Stomata).
 3. Epidermal appendages (ex. Root hairs / Stem hairs).

! EPIDERMIS



- It is outermost protective covering of a plant body.
- It is composed of epidermal cells.
- Epidermal cells → parenchymatous (living)
→ possess large central vacuole due to which cytoplasm becomes peripheral.

≡ Epidermis consists of single layer of cells with no intercellular spaces.

≡ Epidermal cells generally do not possess chloroplast.

NOTE → Cuticle:

≡ It is thick waxy coating above epidermis in leaves, herbaceous stem.

≡ It prevents H_2O loss due to transpiration.

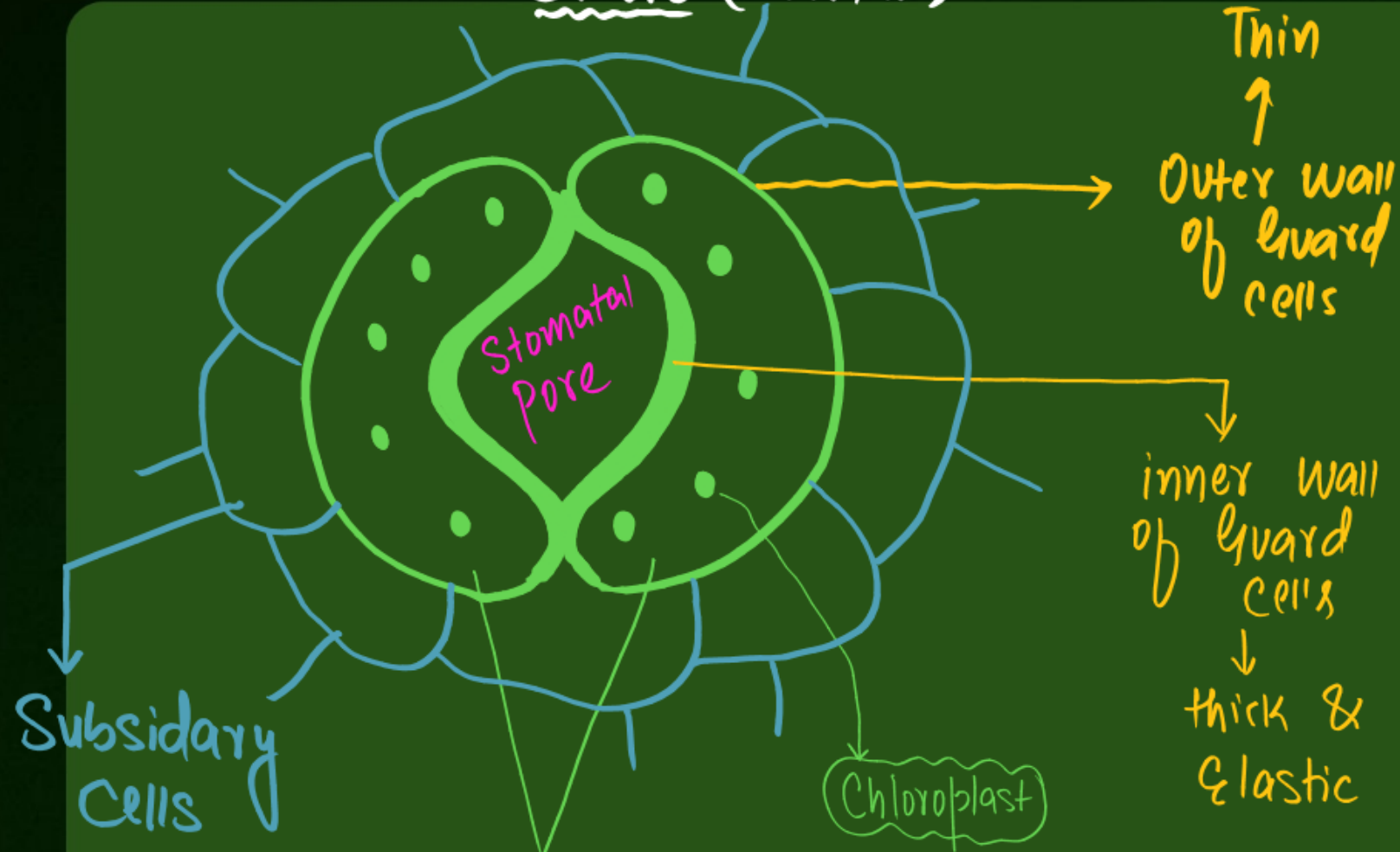
≡ Cuticle is -nt → woody stem.
→ Roots.

2. EPIDERMAL STRUCTURES

- It consists of Stoma apparatus. (present abundantly in the Epidermis of leaves)
- Stomatal apparatus consists of 
 - (i) Two Guard cells
 - (ii) Stomatal pore.
 - (iii) Subsidiary cells — Specialised Epidermal cells (in terms of size and structure).

(Note) → Guard cells are the only Epidermal cells which possess Chloroplast.

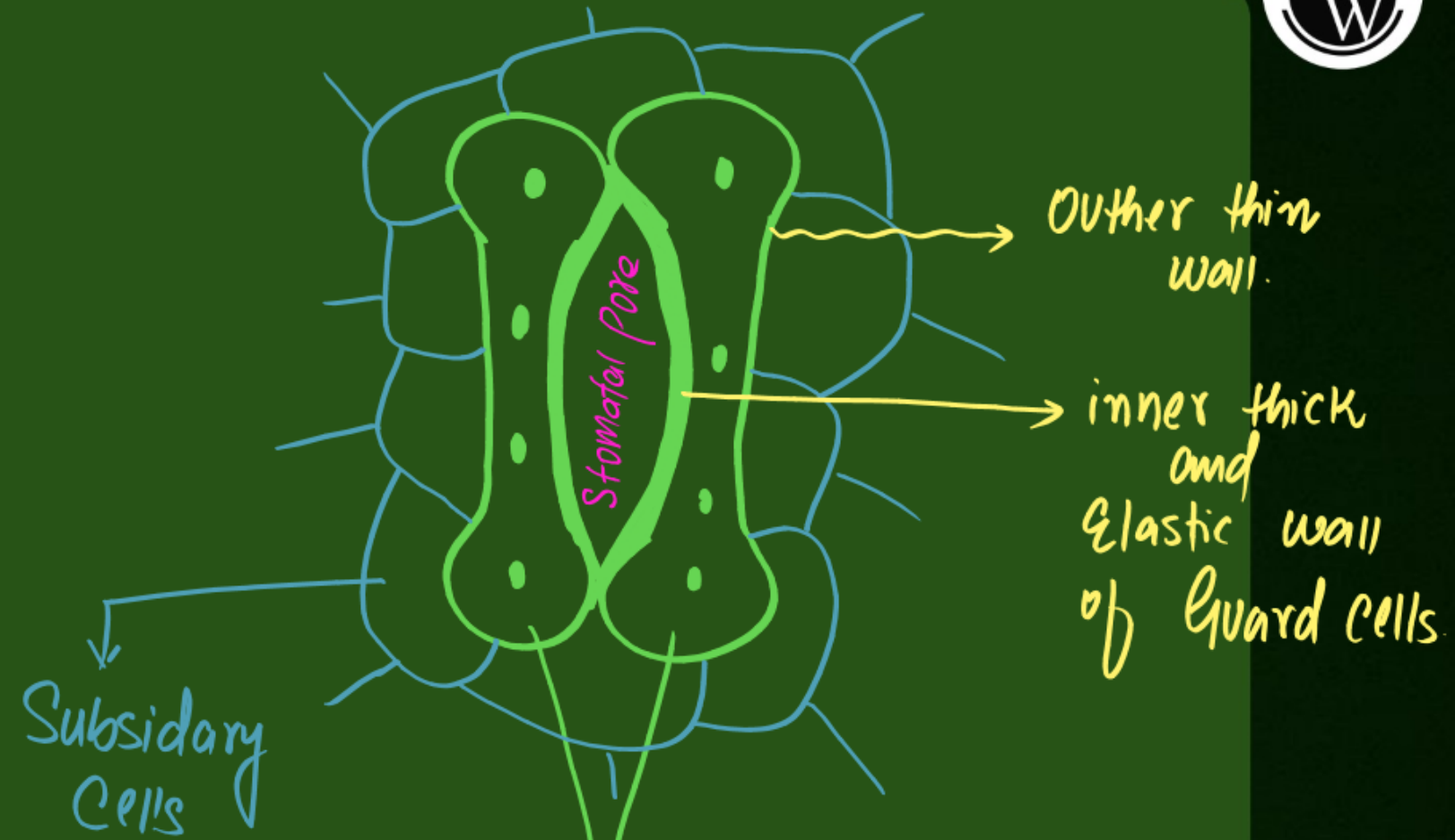
DICOTS (STOMATA)



Two guard cells
(Kidney/Bean Shaped)

↓
Plays an imp. role in
opening and closing of
Stomata.

MONOCOTS (STOMATA)



Two guard
cell
↓
Dumbell Shaped

3. EPIDERMAL APPENDAGES

- It consists of 
 - Root hairs
 - Stem hairs / Trichomes.

Root Hairs

- Arises from epidermis of Root from maturation region.
- Unicellular.
- Increases surface area of root for absorption of H_2O and minerals from soil.
- Root hairs are thin and delicate.



Stem Hairs / Trichomes

- Arises from epidermis of Stem.
- Usually multicellular (may be unicellular also)
- Branched / unbranched.
- Stiff / Soft.
- Prevents H_2O loss due to transpiration.

QUESTIONS

The apical meristem of the root is present:

(2003)

- A** Only in radicals
- B** Only in tap roots
- C** Only in adventitious roots
- D** In all the roots

QUESTIONS

Which meristem helps in increasing girth?
width.

(1998)

- A** Lateral meristem
- B** Intercalary meristem
- C** Primary meristem
- D** Apical meristem

QUESTIONS

H/W

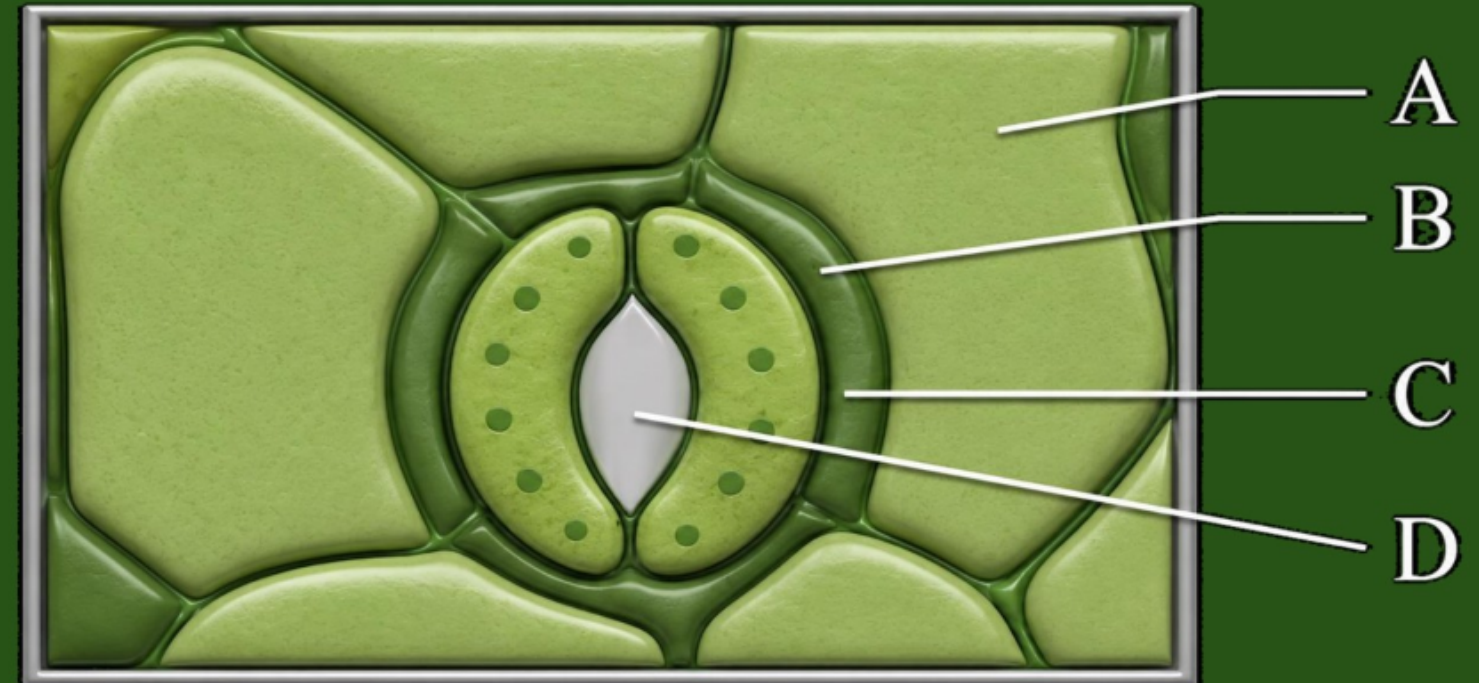
In the given figure, which component has thin outer walls and highly thickened inner walls?
[2024]

A A

B B

C C

D D



HOMEWORK



1 Question

Thank
You